Trigonometric identities

1 Express $\tan x$ in terms of $\cos x$ and $\sin x$.

2 Find the value of $\cos x$ given the following:

a $\sin x = -\dfrac{\sqrt{3}}{2}$ and $\tan x = -\sqrt{3}$

b $\sin x = \dfrac{b}{c}$ and $\tan x = \dfrac{b}{a}$

3 Find the value of $\sin x$ given the following:

a $\cos x = \dfrac{15}{17}$ and $\tan x = \dfrac{8}{15}$

b $\cos x = -\dfrac{5}{13}$ and $\tan x > 0$

4 Find the value of $\tan x$ given the following:

a $\sin x = \dfrac{4}{5}$ and $\cos x = \dfrac{3}{5}$

b $\sin x = \dfrac{2}{3}$ and $\cos x < 0$

5 Write $\csc \theta$ in terms of $\sin \theta$.

6 Simplify the following expressions:

a $\dfrac{\cos \theta}{\sin \theta}$

b $\dfrac{\sin \theta}{\csc \theta}$

c $\dfrac{\sec x}{\cos x}$

d $\dfrac{\csc \theta}{\sec \theta}$

e $\sin \theta \sec (\theta)$

f $\csc \theta \tan \theta \cos \theta$

g $\tan \theta \cos \theta$

h $\tan \theta \cot \theta$

i $\sin (90^\circ - y) \tan y$

j $\dfrac{\sin \theta - \cos \theta}{\cos \theta}$

k $\dfrac{\tan \theta}{\csc \theta \sec \theta}$

l $\sec \theta \cot \theta$

m $\csc \theta \tan \theta$

7 Find the exact value of $\tan x$ given the following equations:

a $\sin x = 2 \cos x$

b $17 \cos x - 31 \sin x = 0$

c $6 \sin x - 11 \cos x = 0$

d $2 \sin^2 x - \cos^2 x = 0$

8 Find $\sec \theta$ if $\cos \theta = \dfrac{2}{\text{[illegible]}}$.

9 Find $\csc\theta$ if $\sin\theta = \frac{4}{7}$.

10 Find $\cot\theta$ if $\tan\theta = 11$.

11 Find the value of $\cos\theta$ if $\sec\theta = \frac{8}{3}$.

12 Find the exact value of $\sin\theta$ if:

a $\csc\theta = \sqrt{3}$

b $\csc\theta = -\frac{2\sqrt{6}}{3}$

13 Consider the following statement about angle θ:
$\tan\theta = 3.5$ and $\cot\theta = -3.5$
Is this statement possible or impossible? Explain your answer.

Pythagorean identities

14 State whether the following statements are correct:

a $\sin^2\theta + \cos^2\theta = 1$

b $\tan^2 x + 1 = \sec^2 x$

c $\sin^2\theta + \cos^2\theta = 2$

d $1 + \cot^2\theta = \sec^2\theta$

15 Consider the statement: $\sin^2(\theta) + \cos^2(\theta) = 2$
Is the statement possible or impossible?

16 Simplify the following expressions:

a $\tan\theta\cos^2(\theta)\sec\theta$

b $(\cos\theta - 1)(\cos\theta + 1)$

c $(\cos\theta - \sin\theta)^2$

d $\cos\theta\sin^2(\theta) - \cos\theta$

e $(3 - \cos x)^2 + \sin^2(x)$

f $\left(1 + \tan^2(u)\right)\left(1 - \sin^2(u)\right)$

g $\left(1 - \sec^2(\theta)\right)\cot\theta$

h $\csc^2(\theta) - \cot^2(\theta)$

i $\cos\theta(\sec\theta - \cos\theta)$

j $(1 - \csc\theta)(1 + \sin\theta)$

k $\sec^2(x)\left(\cos^2(x) - 1\right)$

l $(\sec\theta - \csc\theta)(\cos\theta + \sin\theta)$

17 Simplify the following expressions:

a $\dfrac{\sin^2(\theta)}{1-\sin^2(\theta)}$

b $\dfrac{1-\sin^2(\theta)}{\sin^2(\theta)+\cos^2(\theta)}$

c $\dfrac{1}{1+\tan^2(x)}$

d $\dfrac{\sec^2(\theta)}{\tan^2(\theta)}$

e $\dfrac{\csc^2(\theta)}{\cot^2(\theta)}$

f $\dfrac{\sec^2(x)-1}{\csc^2(x)-1}$

g $\dfrac{1}{\csc^2(\theta)-1}$

18 Simplify the following expressions:

a $\dfrac{1}{1-\cos\theta}+\dfrac{1}{1+\cos\theta}$

b $\dfrac{1}{1-\cos^2(x)}-1$

c $\dfrac{1}{1-\sin^2(x)}-1$

d $\tan\theta+\dfrac{1}{\tan\theta}$

e $\dfrac{1}{1-\sin x}\times\dfrac{1}{1+\sin x}$

f $\dfrac{1}{1-\cos^2(x)}-1$

g $\dfrac{\sin\theta}{1+\cos\theta}+\dfrac{1+\cos\theta}{\sin\theta}$

h $\dfrac{1}{\csc^2(\theta)}+\dfrac{\cos\theta}{\sec\theta}$

i $\dfrac{1}{\cos^2(x)}-\dfrac{1}{\cot^2(x)}$

j $\dfrac{1}{1+\sec\theta}-\dfrac{1}{1-\sec\theta}$

k $\dfrac{\sec x}{\tan x}-\dfrac{\tan x}{\sec x-1}$

l $\dfrac{\sec\theta}{\tan\theta}-\dfrac{\tan\theta}{\sec\theta}$

19 Simplify $\sqrt{a^2+x^2}$, where $x=a\tan\theta$, a is a constant, and $0^\circ<\theta<90^\circ$.

20 Write an identity that expresses $\cot x$ in terms of only the trigonometric ratio $\csc x$.

21 Given that $\cos x=\dfrac{12}{13}$ where x is in the first quadrant, find the exact value of:

a $\sin x$

b $\tan x$

22 Given that $\sin\theta=\dfrac{\sqrt{3}}{2}$ for $90^\circ<\theta<180^\circ$, find the value of $\cos\theta$ using the Pythagorean identity.